



UNIVERSIDADE DE COIMBRA
FACULDADE DE CIÊNCIAS E TECNOLOGIA
DEPARTAMENTO DE ENGENHARIA INFORMÁTICA

NATURAL LANGUAGE INTERACTION 2026/2027

MASTER IN ARTIFICIAL INTELLIGENCE

Course Project

Conversational Paradigms

Professors:

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1 Introduction

The course project of Natural Language Interaction (ILN) has the main goal of exploring at least two different approaches for creating a conversational agent. It should be developed in **groups of two students** and resort to the Python programming language for experimentation and performing experiments. Whenever possible, suitable libraries available in this language should be used.

2 Deliverables

The project has three deliverables to be submitted and presented to one professor during specific dedicated classes. The main result of the first two (D1, D2) is a conversational agent for each. For the third, groups must select **one of two** significantly different options (D3-O1, D3-O2).

D1 Rule-based agent

- * Consists of the adaptation of an agent available from NLTK¹ to a new persona, which may communicate in a different language.
- * Definition of the covered interactions is up to the group.
- * Regular expressions will be the starting point for rules, but other NLP tools can also be explored for different purposes.

D2 Expert **Retrieval-Augmented Generation** (RAG) agent

- * Based on an open Large Language Model (LLM) and on a Vector Database on a selected domain of expertise, created with data collected for the purpose.
- * The agent is expected to provide useful answers to questions in the domain of expertise.
- * Quantitative comparison of answer quality with and without RAG is required.

D3-O1 Integration of **agency** and **detailed evaluation**

- * Requires the integration of the RAG agent with MCP servers useful for the domain or which complement the agent features coherently; as well as a detailed evaluation of the solution,

¹<https://www.nltk.org/api/nltk.chat.html>

following common practices and metrics for RAG, established in the literature, e.g., resorting to LLM-as-Judge.

D3-O2 Research work on the risks of LLMs

- * Study a known risk based on available literature, define it, explore its impact / presence in two or more LLMs, with or without RAG, starting with the one used in D2, and collect examples.
- * Possible risks to study: gender / racial / religious bias, antropomorphisation, inconsistency, sycophancy, plausibility vs faithfulness, memorisation, jailbreak vulnerability, false reasoning, prompt injection, etc.

Ideally, the agents of D1 and D2 will be **integrated in a single system** where, depending on the interaction, a different agent will be triggered for handling it, e.g., the rule-based agent may handle chit-chat, whereas the expert handles domain-specific questions. A text classifier may be developed for this purpose.

Besides the creation of the conversational system, all the complementary work will be rewarded. This includes tasks like data creation and preparation, comparison of approaches and evaluation, quality of the conclusions, clarity of the final report and presentation, as well as the integration of other concepts learned in the theoretical classes, if well-supported.

3 Documentation

D1 and D2 do not require the formal submission of a report, but **must be presented to and discussed with one of the professors**, during a specific class. **D3 entails the submission of a report compiling the three deliverables**, to be presented and defended orally, to both professors, in the end of the course.

This report should have **at most six pages**, excluding references, and be formatted according to IJCAI's style². The following structure is suggested:

1. Introduction, including motivation and objectives;
2. Detailed description of the implemented agents, including options taken, used data, and explored approaches;

²https://www.ijcai.org/authors_kit

Attention: plagiarism will not be tolerated!

3. Experimental validation, including usage examples, problematic cases and their discussion, and the analysis / evaluation of components of the work that support the technological choices;
4. Detailed evaluation **OR** studied risk;
5. Conclusion and main takeaways, challenges and limitations;
6. Bibliographic references.

4 Important Dates

The deliverables must be submitted to Inforestudante in the following dates:

- D1** submitted on Tuesday, **22nd September**, presented on Wednesday, **23rd September**
- D2** submitted on Tuesday, **29th September**, presented on Wednesday, **30th September**
- D3** submitted on Tuesday, **6th October**, presented on Thursday, **8th October**

5 Evaluation Criteria

The NLI project has a weight of **60% of the final evaluation** of the course (i.e., **12 points in 20**), distributed as follows:

- D1** 2 points.
- D2** 3 points.
- D3** including the report and the presentation, 7 points.

Evaluation will consider the following main aspects:

- Adherence to the main requirements;
- Creativity and complexity of the developed agents;
- Integration of other concepts learned in the theoretical classes;
- Experimental validation;
- Quality of the documentation, conclusions and discussion;
- Final presentation and oral discussion of the work.